



Reducing Bias in Cluster Randomized Trials in Nephrology

Jessica Kasza ¹, Gabor Mihala ^{2,3}, Agnès Caille ^{4,5} and Rhys Bowden ¹

JASN 37: 2055–2058, 2026. doi: <https://doi.org/10.1681/ASN.0000001189>

Cluster Randomized Trials in Nephrology

Cluster randomized trials are an important class of randomized controlled trials. Instead of randomizing individual participants, cluster randomized trials randomize entire groups (clusters) of participants. These trials are useful when assessing the effects of interventions applied at a cluster level on participant-level outcomes and can be used to assess participant-level interventions. There are many different variants of cluster randomized designs, including stepped wedge designs. Here, we consider standard cluster randomized trials: clusters are randomized to implement one of two treatments.¹

Cluster randomized trials are common in nephrology. For example, TEACH-PD randomized dialysis facilities to a new training curriculum for new peritoneal dialysis patients or to continue with their usual approach, with the primary outcome as time to first peritoneal dialysis-related infection.² The HiLo trial randomized facilities to target a Hi or Lo serum phosphate concentration (≥ 6.5 versus < 5.5 mg/dl) for all patients at that facility, comparing mortality and hospitalizations. Interestingly, HiLo switched to an individually randomized design mid-trial because of a risk of biased patient recruitment.³

Indeed, cluster randomized trials are vulnerable to biases and to a specific form of bias.⁴ If a trial has been subject to bias, any observed differences/similarities in outcomes between treatment groups may not only be due to treatments and randomness. Throughout a trial—from planning to analysis and reporting—trial teams should consider ways in which biases may arise. Preventing bias entirely may be impossible, but by identifying how biases could arise, trial teams can aim to reduce their influence and evaluate plausible directions of biases.

Biases in Cluster Randomized Trials

There are several ways in which bias may arise during trials. The Risk of Bias 2 (RoB 2) tool provides a framework for considering the domains of bias that trials may be subject to.⁴ These are biases arising from the randomization process; because of deviations from treatments; induced by

missing data; in outcome measurement; and because of selecting reported results.

Cluster randomized trials may be particularly vulnerable to bias because of cluster-level application of treatments with blinding rarely feasible and recruitment of participants within clusters.⁴ There is a tendency to randomize few clusters, further exacerbating risks of chance imbalances.⁵ Some clusters may drop out of trials or recruit/include no participants. Postrandomization cluster withdrawal is problematic because the intention-to-treat principle implies that all randomized clusters must be included in analysis. Furthermore, clusters may merge/split, and participants could switch between clusters.⁶

An additional bias domain specific to cluster randomized trials arises from identification/recruitment of participants within clusters.⁴ If participants are identified/included after their cluster is randomized, and those who identify/recruit participants (or participants themselves) are aware of the allocation, there is a risk of cluster allocation influencing which participants are included, potentially leading to imbalanced treatment groups.

Minimizing the Risk of Bias in Cluster Randomized Trials

The standard approaches to minimizing bias in individually randomized trials can be applied to cluster randomized trials, *e.g.*, ensuring randomization is conducted appropriately, treatments delivered as described, data collection procedures guard against bias, and publishing a detailed statistical analysis plan before analysis. Figure 1A provides short descriptions of how bias could arise in cluster randomized trials in nephrology and how risks of bias were/could have been mitigated in the TEACH-PD or HiLo trial.

If a cluster randomized trial includes few clusters, even if randomization has been conducted appropriately, imbalances of cluster-level characteristics between groups may arise by chance. Using stratified randomization, matching, minimization, or covariate-constrained approaches to ensure balance on key cluster-level characteristics is particularly important.⁷ To reduce bias when estimating treatment

¹School of Public Health and Preventive Medicine, Monash University, Melbourne, Victoria, Australia

²Centre for Health Services Research, The University of Queensland, Woolloongabba, Queensland, Australia

³Australasian Kidney Trials Network, The University of Queensland, Woolloongabba, Queensland, Australia

⁴Université de Tours, Université de Nantes, INSERM, Tours, France

⁵Inserm CIC 1415, CHRU de Tours, Tours, France

Correspondence: Prof. Jessica Kasza, email: jessica.kasza@monash.edu

Published Online Ahead of Print: June 17, 2026

A	Bias domain	Example of how this risk of bias could arise in nephrology cluster randomized trials	How this risk of bias was mitigated in TEACH-PD or HiLo and any suggestions for improvements
	Bias arising from randomization process	If there is no allocation concealment of clusters or the sequence was not randomly generated, bias can arise	TEACH-PD: Dialysis centers were randomized using an allocation sequence generated by an independent, unblinded statistician, which was only accessible to that person; a low risk of bias
	Bias from identification or recruitment of participants within clusters	If patients are recruited after their site was randomized and there is no blinding, recruiters may preferentially approach those patients they believe will benefit from the site's intervention, instead of all eligible patients	HiLo: The Timeline Cluster diagram for the HiLo trial (Panel B below) indicates a lack of blinded recruitment, and baseline serum concentrations were different in patients recruited in different arms. There was a high risk of bias. The potential for this risk could have been mitigated by recruiting participants with blinded staff
	Bias due to deviations from intended intervention	If participants and/or those delivering the intervention at a site are aware of the site's assigned intervention, this may lead to nonadherence, use of additional interventions, or the intervention not being delivered as planned	HiLo: Clinicians delivering the intervention were aware of their site's intervention. To help ensure that concentrations were achieved as specified, sites were monitored remotely, receiving monthly reports. This helped ensure consistency of the interventions between and within sites over time. A risk of bias exists but efforts were made to reduce the risk
	Bias due to missing outcome data	If data are missing more often in one arm (e.g., due to less contact with patients), there is risk of bias. Bias may arise if outcome missingness is related to its true value	HiLo: To reduce missing data, case report forms were replaced with a portal for sites to upload electronic health records; hence, data collection was identical across sites and arms. It seems unlikely that missingness was related to outcome values. There appears to be a low risk of bias
	Bias in measurement of the outcome	Reporting of outcomes could be influenced if participants or outcome assessors are aware of the cluster's allocation	TEACH-PD: Outcomes were collected in "real time" in a national clinical registry as per local standard practice. There thus appears to be a low risk of bias
	Bias in selection of the reported results	Including different variables in outcome regression models may lead to different conclusions. Authors may be tempted to report the most favorable estimate	TEACH-PD: The statistical analysis plan was posted online prior to analysis (Mihala <i>et al.</i> , https://doi.org/10.48610/00beb28). Readers can thus affirm that reported analyses were prespecified, and all prespecified analyses are reported. A low risk of bias would be anticipated

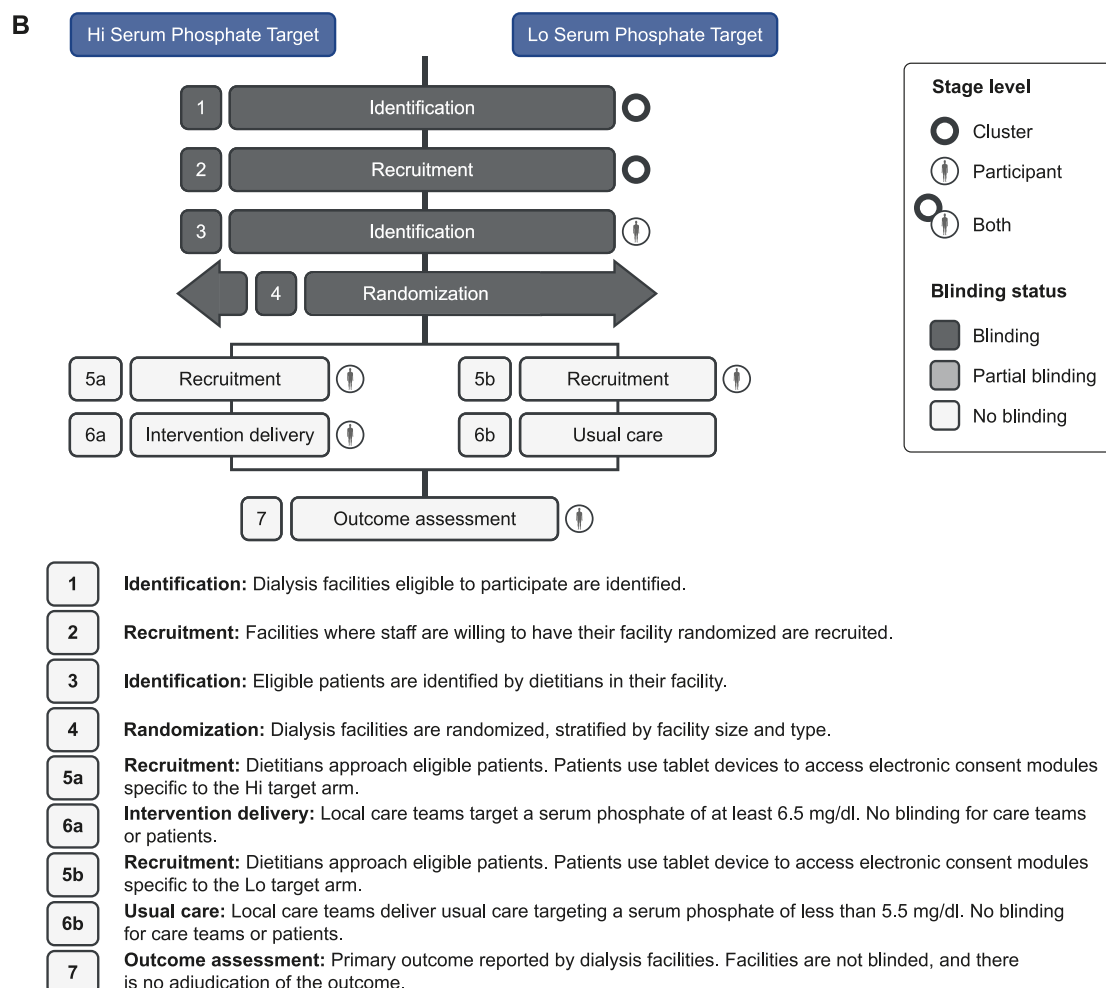


Figure 1. Examining domains of bias in TEACH-PD and HiLo, two cluster randomized trials in nephrology. (A) Domains of bias, brief summaries of how these risks of bias could arise in nephrology research, and examples of how these risks of bias were mitigated in the TEACH-PD or HiLo trials. (B) The Timeline Cluster diagram for the HiLo cluster randomized trial. This diagram is based on the information available in the HiLo trial protocol and report and as such is *post hoc* and for illustrative purposes only.

effects, researchers should prespecify methods for adjusting for characteristics used in randomization and those predictive of the outcome, missingness, or differential recruitment.⁸

The potential for cluster-level drop-out and deviations from assigned treatments, and of the biases that may result, can be minimized by ensuring that clusters are aware of the possibility of randomization to either treatment and clarifying expectations regarding engagement with allocated treatments. Reducing risks of cluster splits/mergers may not be possible, but to reduce the associated risks of bias, researchers should prespecify how data from split/merged clusters will be analyzed.

The risk of bias because of identification/recruitment of participants within clusters arises when participants are identified/recruited after cluster randomization. If participants are recruited by people aware of their cluster's allocation, there is potential for bias, *i.e.*, there could be differential selection and/or consent rates in the two groups with differences in the number of participants included per treatment group and also imbalance in participant-level characteristics. To minimize risks of bias because of these differential timelines, eligible participants should be identified before clusters are randomized, which may be difficult/impossible. For example, if individual consent is required,⁹ recruitment of participants may only be possible after a cluster has been randomized. In this case, recruiters and potential participants should be blinded to the cluster's allocation and the consent process identical regardless of allocation wherever possible.

To identify the risk of this bias domain, it is recommended that the timing of cluster randomization and cluster and participant identification and recruitment as well as who is unblinded to cluster allocation at each stage be carefully described, ideally before recruitment starts.¹⁰ The information required can be complex, and graphical displays of timing and blinding information can help. The Timeline Cluster is a graphical tool that helps to assess this and other risks of bias.³

The Timeline Cluster and HiLo

HiLo switched from a cluster to an individually randomized trial after 12% of the intended patients were enrolled, because of an imbalance in the baseline serum phosphate levels of patients recruited in each arm.³ It was posited that the imbalance was due to recruitment of patients by facility dietitians after clusters were aware of their allocation. Figure 1B shows the Timeline Cluster diagram for HiLo based on the information available in the published protocol and report. This may not be entirely accurate, *e.g.*, we assume that dietitians' master lists of eligible patients were generated before facility randomization, but these may have been updated throughout the trial.

Figure 1B shows black rectangles at the initial four stages of the trial, indicating no risk of bias because randomization has not yet occurred. When patients were recruited, neither clusters nor patients were blinded (indicated by white boxes): dietitians knew their facility's allocation and consent modules differed between arms.

This indicates the potential for recruitment to differ across arms. This risk would have been reduced if trial teams ensured all eligible patients were approached by dietitians, if patients were recruited by someone external to their facility and blinded to the study arm, or if patients were recruited before facility randomization. However, these steps may have required infeasible increases in the trial duration/cost.

Final Recommendations

The risk of bias can rarely be eliminated in cluster randomized trials but can be minimized through careful planning. The first step is to identify risks of bias, considering each RoB 2 domain and using tools such as the Timeline Cluster. Once risks have been identified, bias mitigation strategies can be developed, weighing costs against risks. If risks of bias are high and researchers are assessing individual-level treatments, they may wish to consider an individually randomized trial instead of a cluster randomized trial.

Disclosures

Disclosure forms, as provided by each author, are available with the online version of the article at <http://links.lww.com/JSN/F909>.

Author Contributions

Conceptualization: Rhys Bowden, Agnès Caille, Jessica Kasza, Gabor Mihala.

Writing – original draft: Jessica Kasza.

Writing – review & editing: Rhys Bowden, Agnès Caille, Jessica Kasza, Gabor Mihala.

Funding

J. Kasza: National Health and Medical Research Council (2033380). R. Bowden: Supported by National Health and Medical Research Council (2037218).

References

- Hayes RJ, Moulton LH. *Cluster Randomised Trials*, 2nd ed. Chapman and Hall/CRC; 2017.
- Chow JSF, Boudville N, Cho Y, et al. Multi-center, pragmatic, cluster-randomized, controlled trial of standardized peritoneal dialysis (PD) training versus usual care on PD-related infections (the TEACH-PD trial): trial protocol. *Trials*. 2023;24(1):730. doi:10.1186/s13063-023-07715-0
- Edmonston D, Isakova T, Dember LM, et al. Higher versus lower phosphate targets for patients undergoing in-center hemodialysis: a randomized controlled trial. *J Am Soc Nephrol*. 2025; 36(12):2445–2455. doi:10.1681/ASN.0000000765
- Eldridge S, Campbell MK, Campbell MJ, et al. *Revised Cochrane Risk of Bias Tool for Randomized Trials (RoB 2): Additional Considerations for Cluster-Randomized Trials (RoB 2 CRT)*. Accessed January 29, 2026 <https://sites.google.com/site/riskofbiastool/welcome/rob-2-0-tool/rob-2-for-cluster-randomized-trials>.
- Hemming K, Thompson J, Kristunas C, et al. The performance of small sample correction methods for controlling type I error when analyzing parallel cluster randomized trials: a systematic review of simulation studies. *J Clin Epidemiol*. 2025;185: 111838. doi:10.1016/j.jclinepi.2025.111838

6. Caille A, Billot L, Kasza J. Practical and methodological challenges when conducting a cluster randomised trial: examples and recommendations. *J Epidemiol Popul Health*. 2024;72(1):202199. doi:[10.1016/j.jepih.2024.202199](https://doi.org/10.1016/j.jepih.2024.202199)
7. Turner EL, Li F, Gallis JA, Prague M, Murray DM. Review of recent methodological developments in group-randomized trials: part 1—design. *Am J Public Health*. 2017;107(6):907–915. doi:[10.2105/AJPH.2017.303706](https://doi.org/10.2105/AJPH.2017.303706)
8. Hemming K, Hall J, Copas A, Watson S, Taljaard M, Hooper R. Covariate adjustment in cluster randomised trials: a practical guide. *BMJ*. 2025;391:e084194. doi:[10.1136/bmj-2025-084194](https://doi.org/10.1136/bmj-2025-084194)
9. Goldstein CE, Weijer C, Taljaard M, et al. Ethical issues in pragmatic cluster-randomized trials in dialysis facilities. *Am J Kidney Dis*. 2019;74(5):659–666. doi:[10.1053/j.ajkd.2019.04.019](https://doi.org/10.1053/j.ajkd.2019.04.019)
10. Caille A, Kerry S, Tavernier E, Leyrat C, Eldridge S, Giraudeau B. Timeline cluster: a graphical tool to identify risk of bias in cluster randomised trials. *BMJ*. 2016;354:i4291. doi:[10.1136/bmj.i4291](https://doi.org/10.1136/bmj.i4291)